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**Morikawa**

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(54) **LIQUID CONTAINER**

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(52) **U.S. Cl.**
CPC ......... ***B41J 2/17513*** (2013.01); ***B41J 2/1752*** (2013.01); *B41J 2002/17516* (2013.01)

(58) **Field of Classification Search**
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See application file for complete search history.

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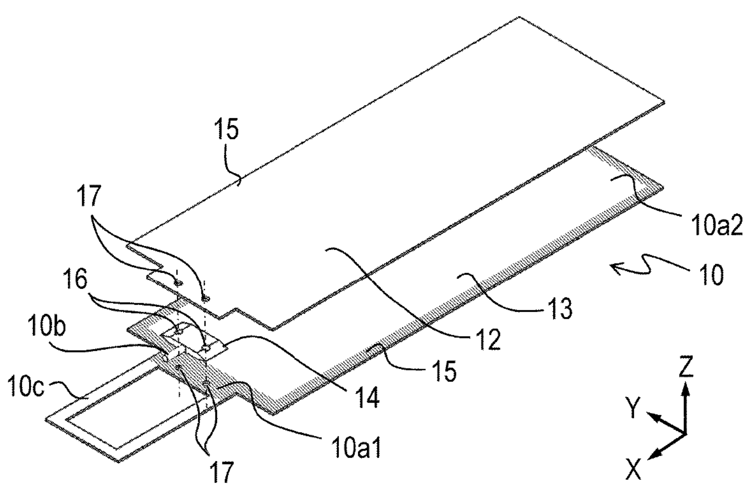
(57) **ABSTRACT**

A liquid container being able to be detachably attached to a liquid ejecting device and supplying a liquid to the liquid ejecting device is provided, the liquid container including: a bag member having an accommodating portion, which accommodates the liquid inside and has flexibility, and a supply port which supplies the liquid in the accommodating portion to the liquid ejecting device; and a positioning member fixed to the bag member and positioned with respect to the liquid ejecting device, wherein the bag member further has a handle portion.

**5 Claims, 9 Drawing Sheets**



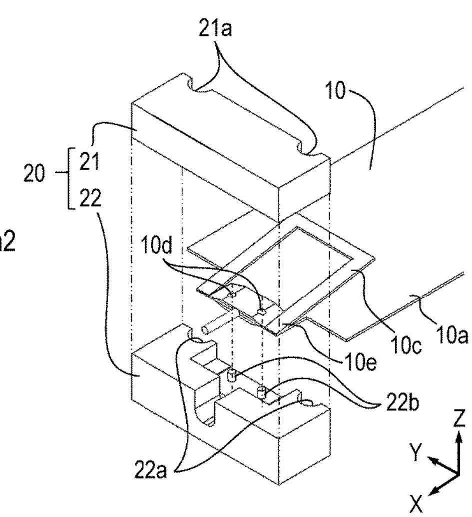
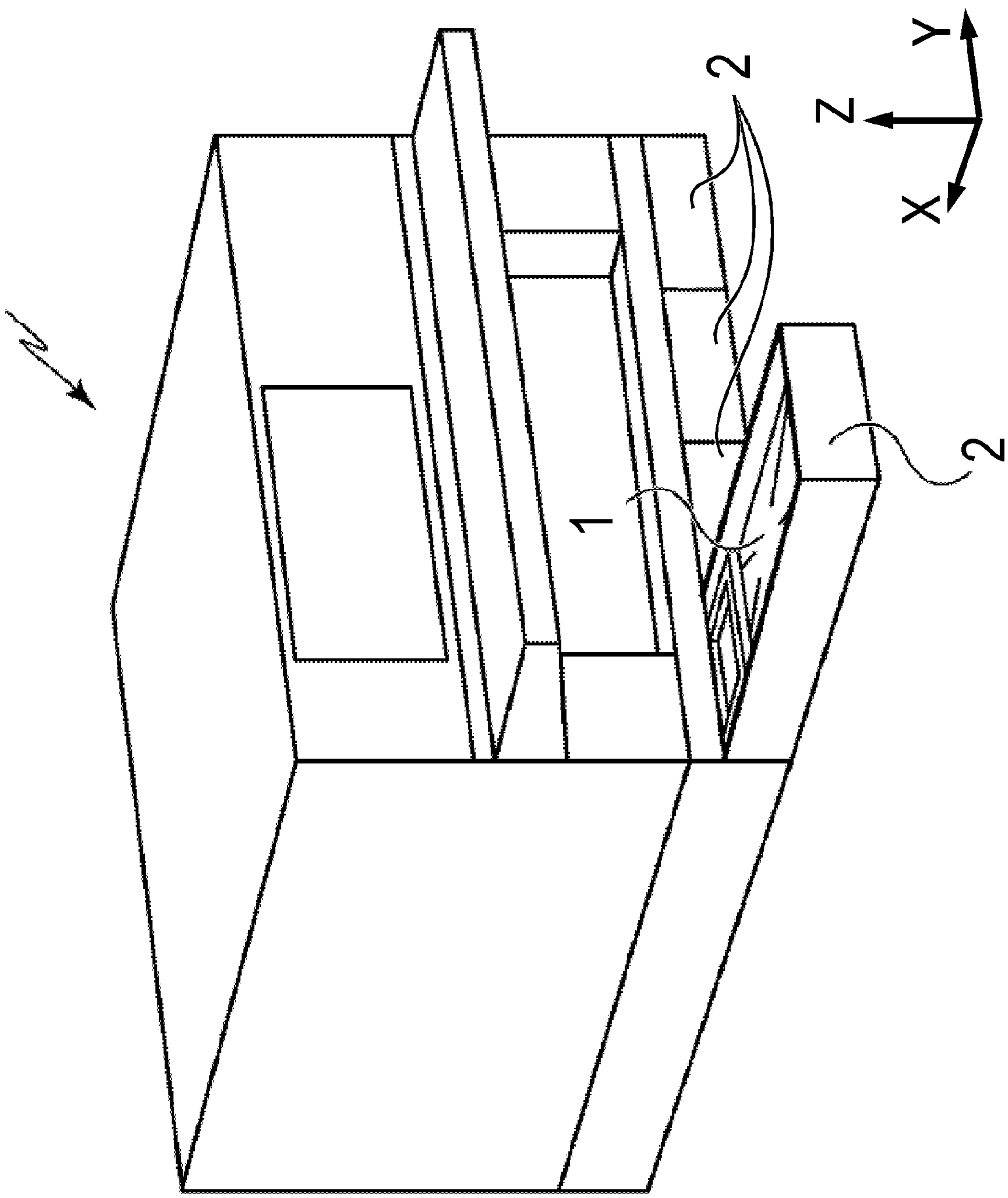



FIG. 1



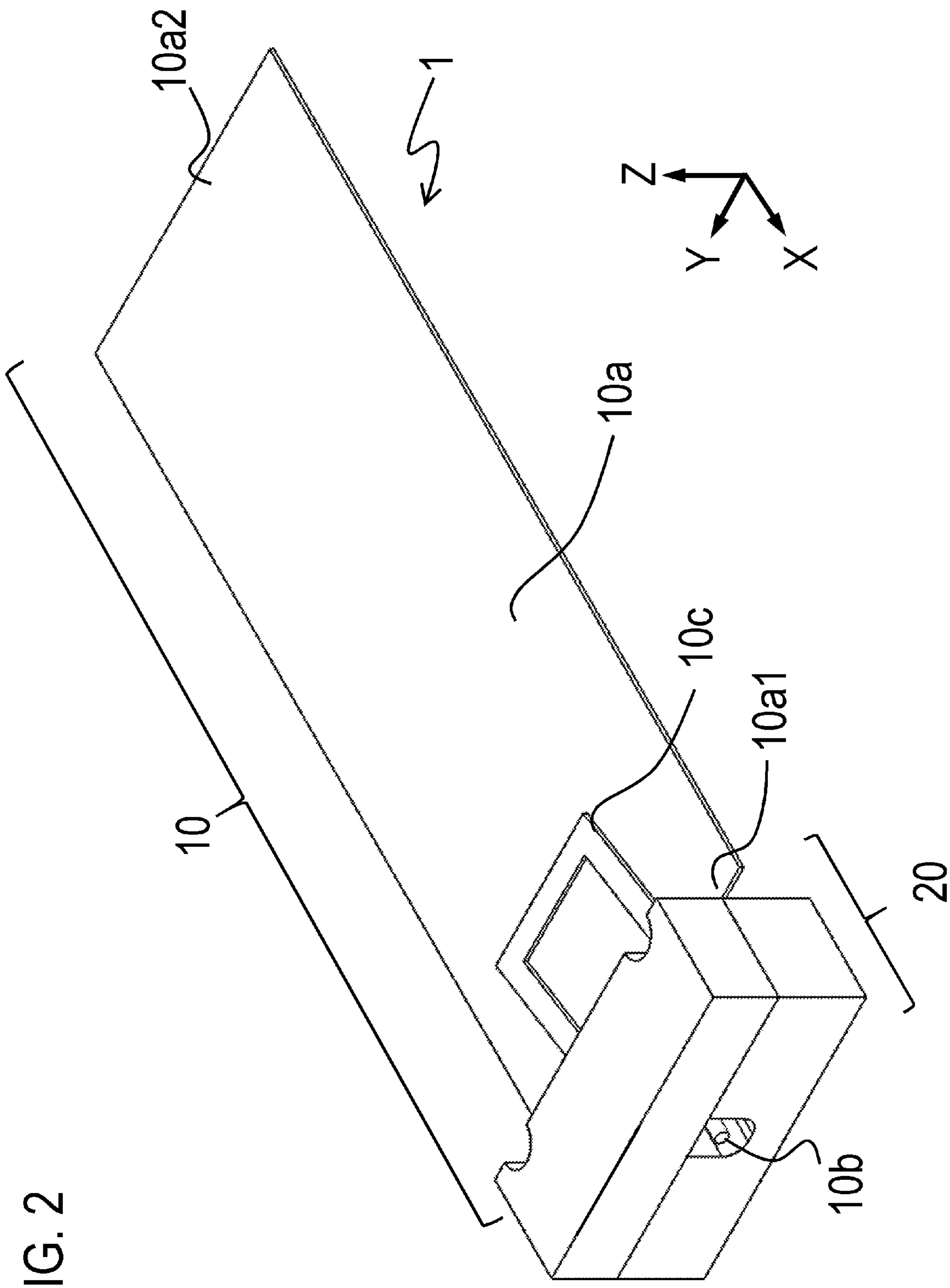



FIG. 2

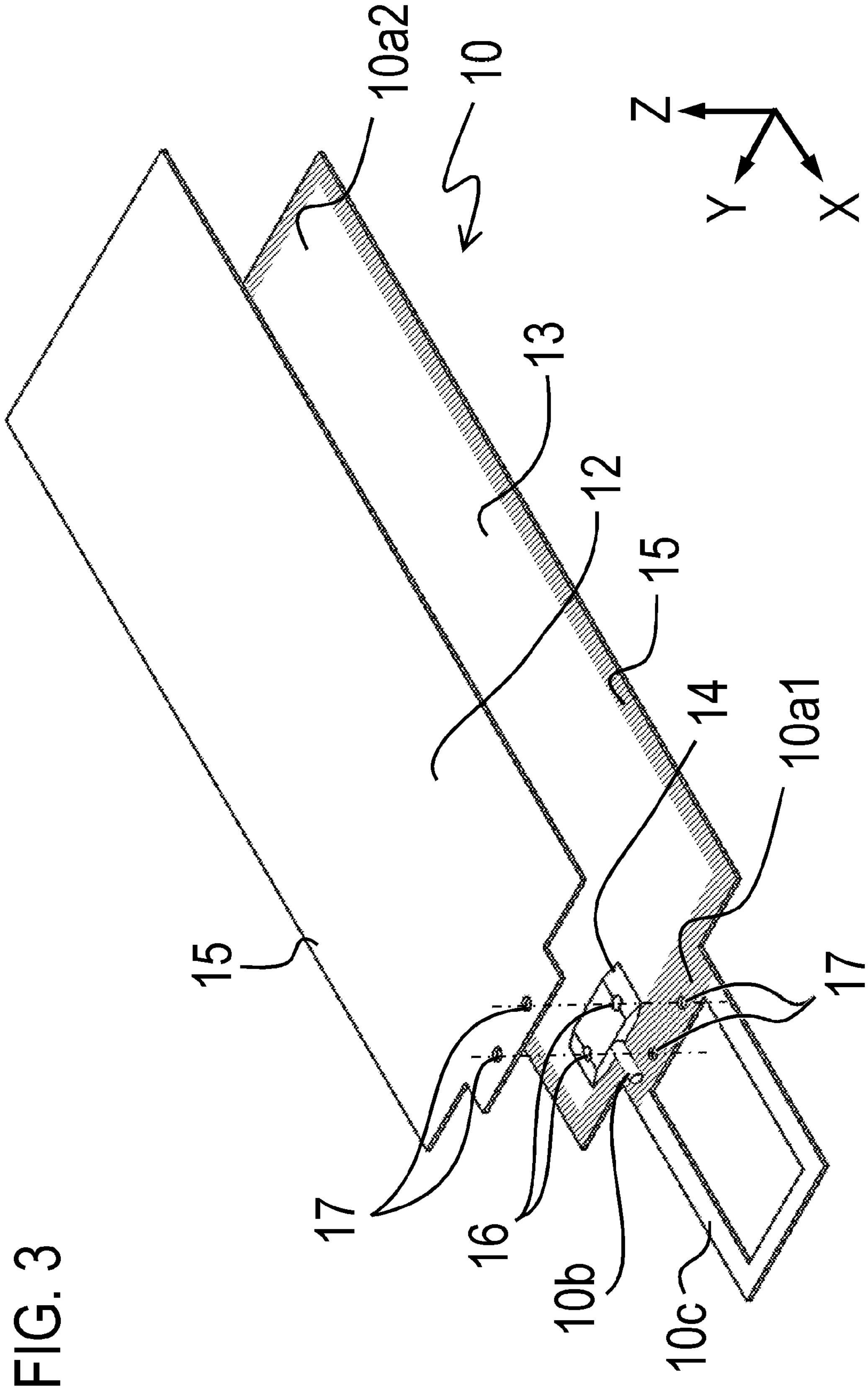



FIG. 3

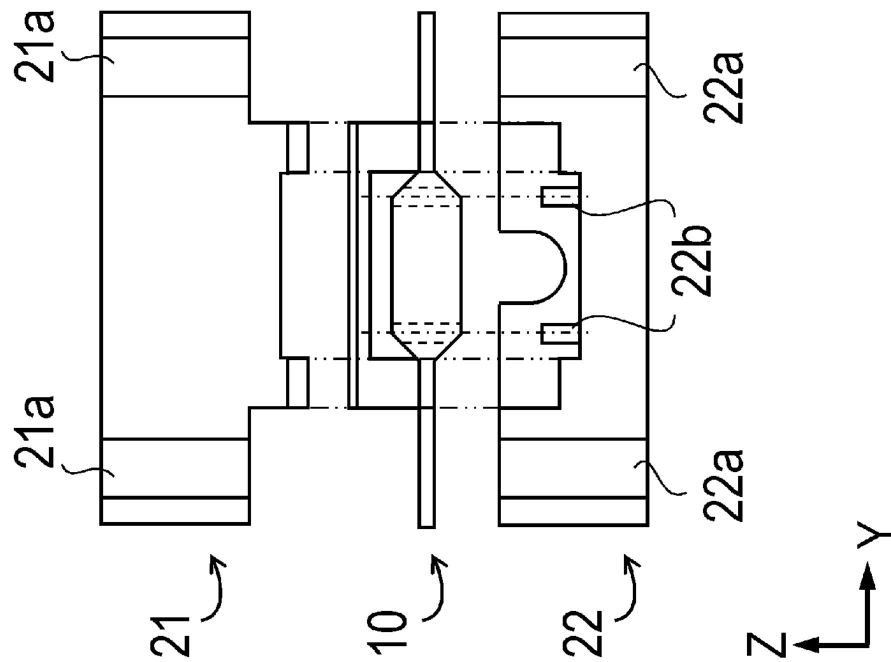

21a
21a
22a
22a
22b
21
10
22
Z
Y

FIG. 4B

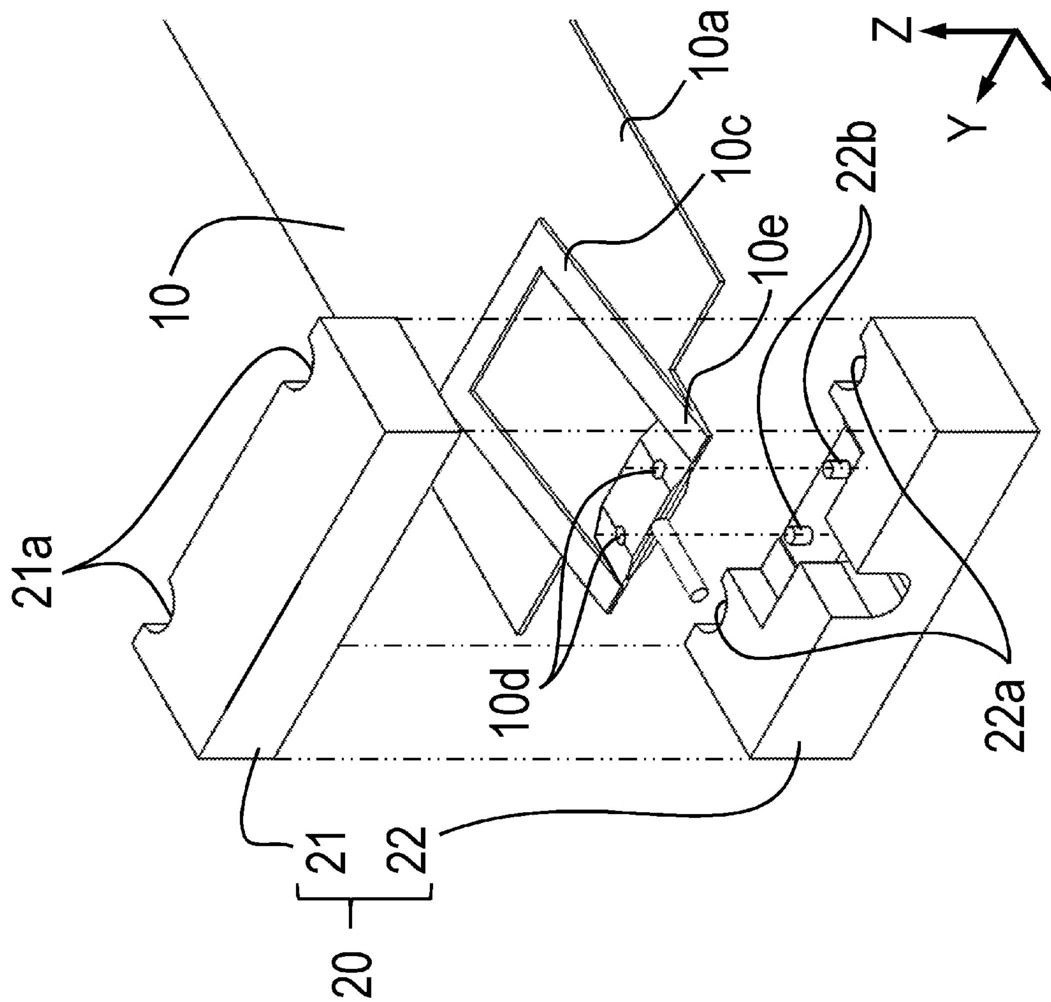

10
10a
10c
10e
22b
21a
10d
22a
21
22
20
Z
Y
X

FIG. 4A

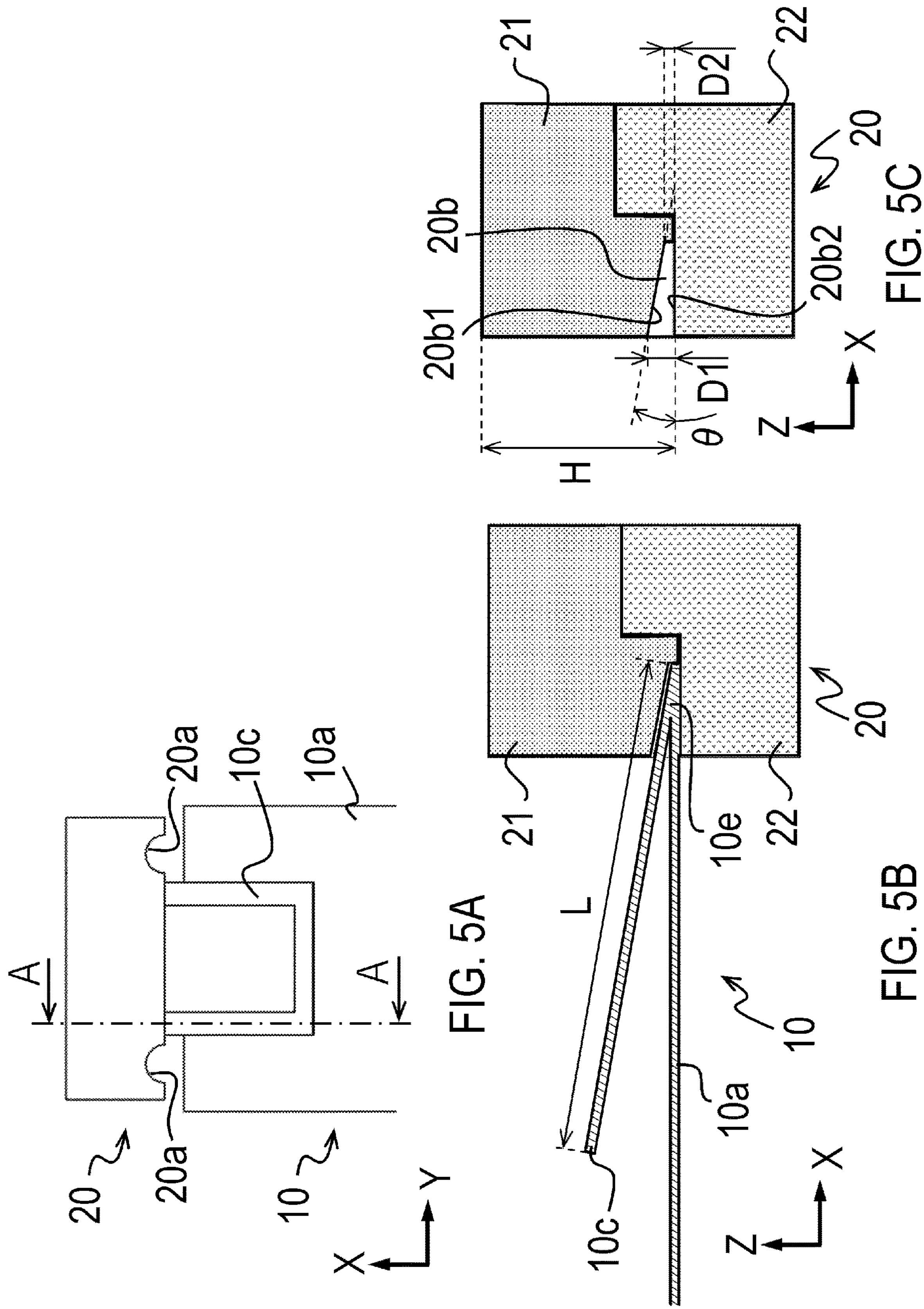



FIG. 5A

FIG. 5B

FIG. 5C

FIG. 6

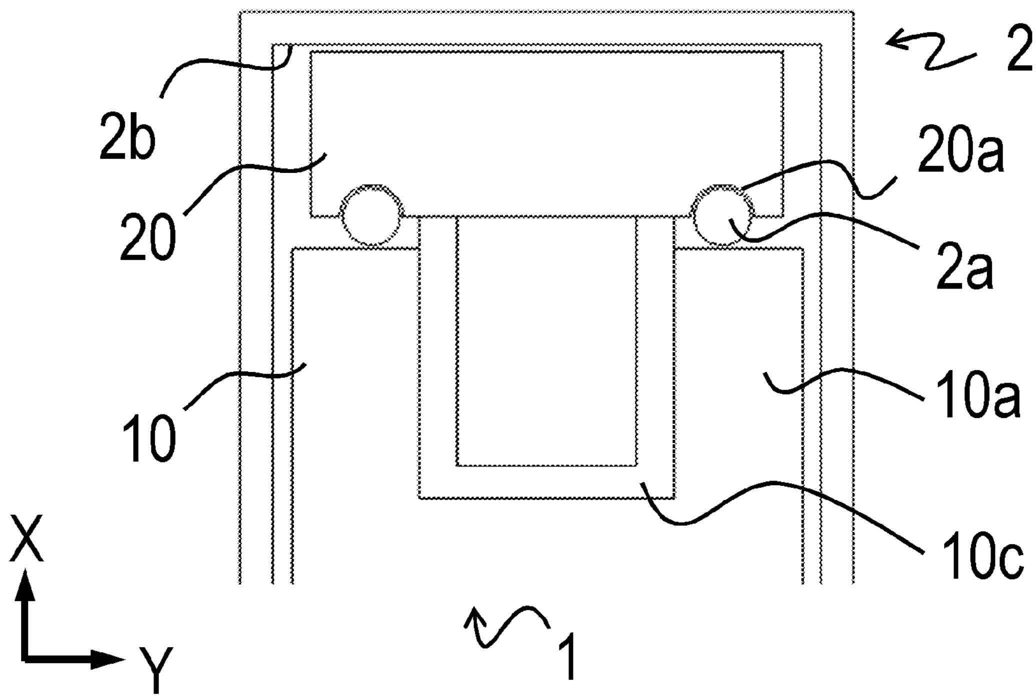
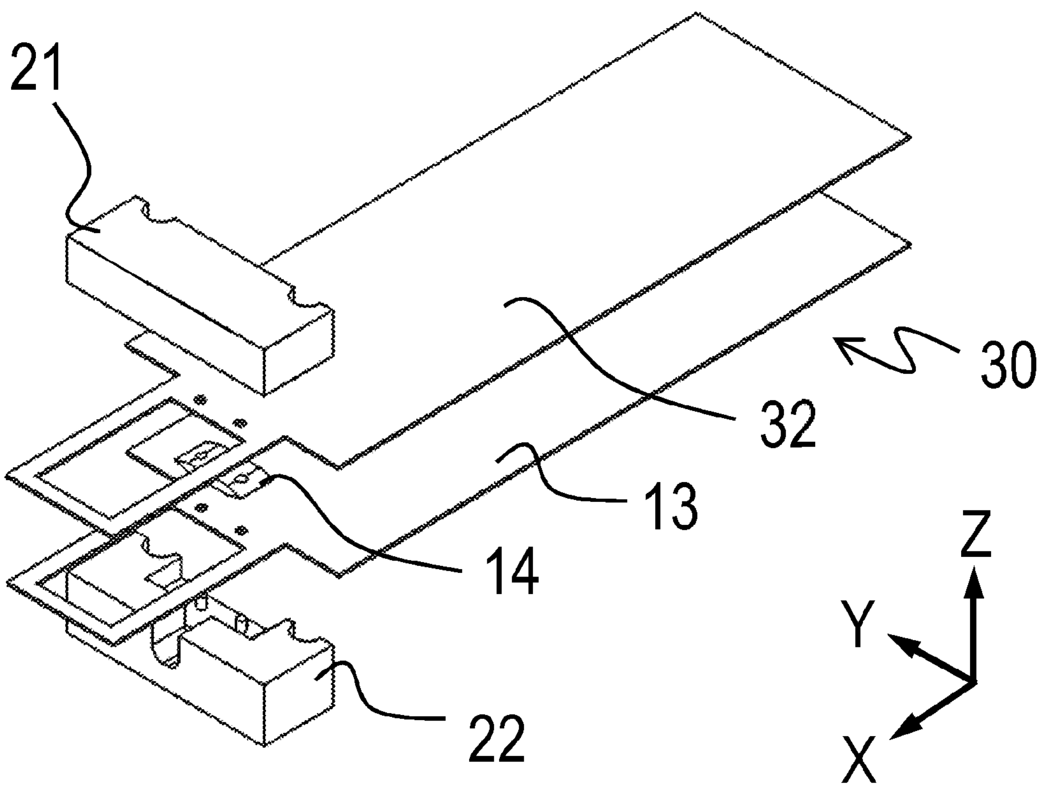


FIG. 7



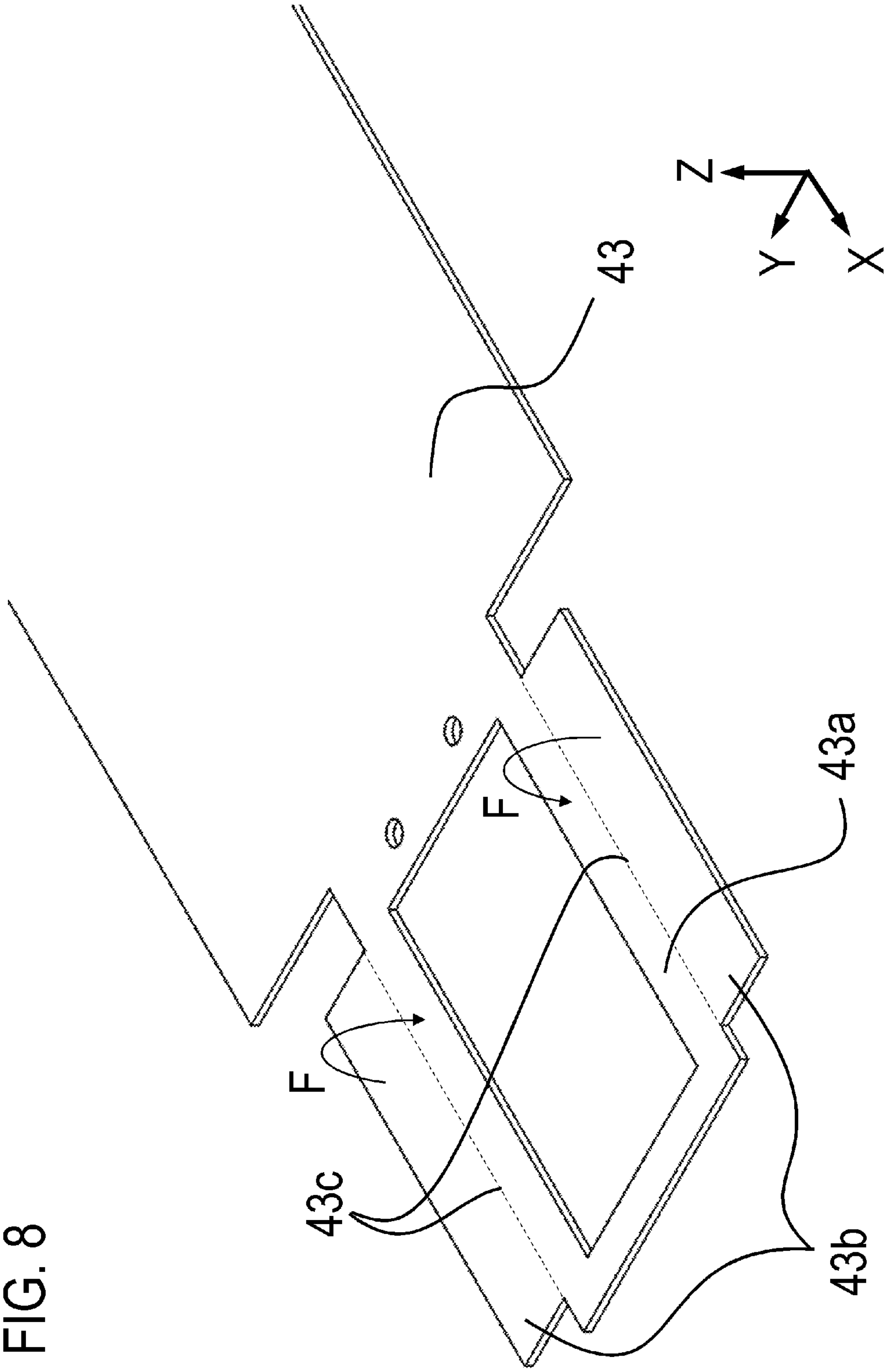



FIG. 8

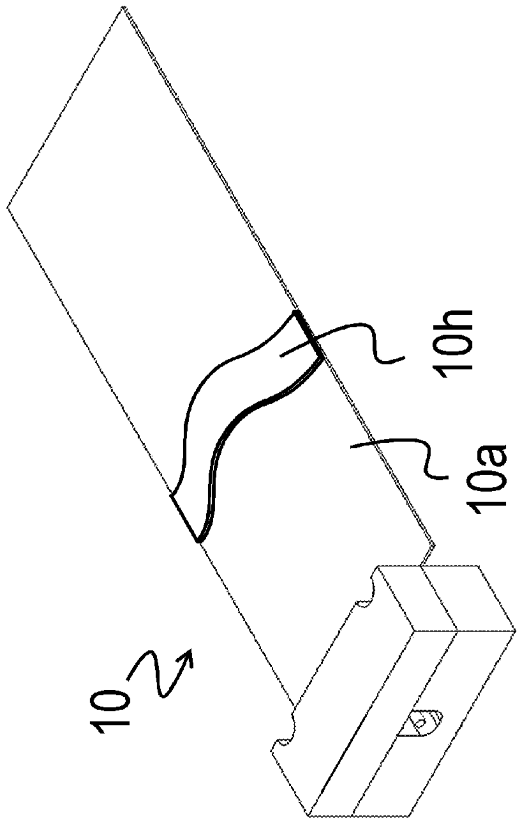



FIG. 9B

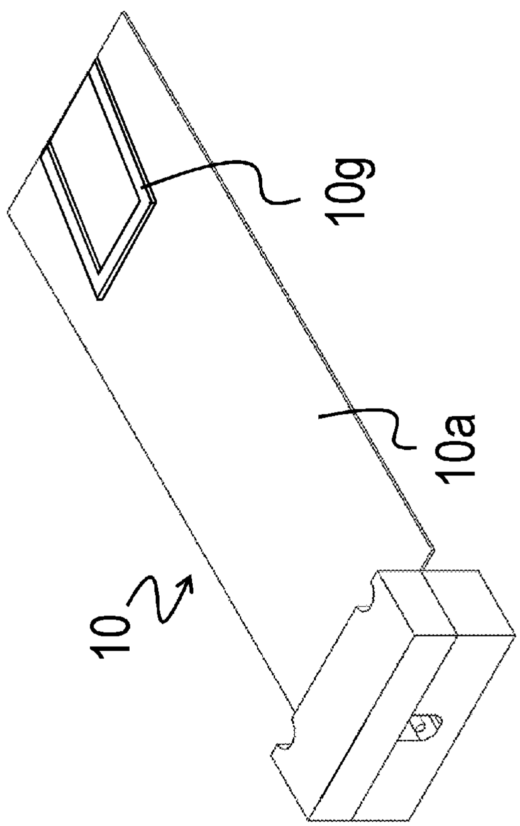



FIG. 9A

# LIQUID CONTAINER

## BACKGROUND OF THE INVENTION

### Field of the Invention

The present invention relates to a liquid container which supplies a liquid to a liquid ejecting device.

### Description of the Related Art

As an aspect of a liquid container which contains a liquid to be supplied to a liquid ejecting device, a so-called ink pack is known (Japanese Patent Application Publication No. 2018-153949, for example). The ink pack contains a liquid such as ink to be supplied to an inkjet printer (hereinafter, also referred to simply as a "printer"), which is an aspect of a liquid ejecting device, in a container such as a bag-shaped member having flexibility.

Some of the printers to which the ink pack is attached include a case in which the ink pack is disposed. In the printer as above, a supply path of the ink between the ink pack and the printer and an electrical communication path are established by disposing the ink pack in a case, such as a cassette, and by attaching the ink pack, together with the case, to the printer.

Japanese Patent Application Publication No. 2018-153949 discloses a constitution of a liquid container including a bag member which accommodates a liquid such as ink and a connecting member connected to the bag member and positioned with respect to the case. Moreover, the connecting member has a handle portion provided in order to improve handling performance when the liquid container is mounted in the case.

## SUMMARY OF THE INVENTION

However, in the constitution in which the handle and the like are provided on the connecting member, the number of components constituting the connecting member increases, and a size of the connecting member becomes larger. Then, in order to accommodate the liquid container in the case, the size of the bag-shaped member needs to be reduced in accordance with the size increase of the connecting member, and accommodation efficiency of the liquid drops. In addition, since the connecting member is constituted by many molded products such as a rotating shaft, a handle portion and the like, manufacturing costs rise.

The present invention has been made in view of the above problems and has an object to provide a liquid container with a simple constitution and high liquid accommodation efficiency.

In order to solve the above problems, the liquid container according to the present invention comprises:

a bag member having an accommodating portion, which accommodates therein the liquid and has flexibility, and a supply port which supplies the liquid in the accommodating portion to the liquid ejecting device; and

a positioning member fixed to the bag member and positioned with respect to the liquid ejecting device, wherein

the bag member further has a handle portion.

Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.



## BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view of a liquid ejecting device in which a liquid container according to a first embodiment is mounted;

FIG. **2** is a perspective view of the liquid container according to the first embodiment;

FIG. **3** is an exploded perspective view of a bag member according to the first embodiment;

FIGS. **4**A and **4**B are exploded views of the liquid container according to the first embodiment;

FIGS. **5**A to **5**C are views illustrating constitutions of a handle portion and a clearance portion according to the first embodiment;

FIG. **6** is a top view of a state in which the liquid container according to the first embodiment is mounted on a cassette;

FIG. **7** is an exploded perspective view of a liquid container according to a second embodiment;

FIG. **8** is a perspective view illustrating a constitution of a sheet-state member according to a third embodiment; and

FIGS. **9**A and **9**B are perspective views of a liquid container according to a variation.

## DESCRIPTION OF THE EMBODIMENTS

Hereinafter, a description will be given, with reference to the drawings, of embodiments (examples) of the present invention. However, the sizes, materials, shapes, their relative arrangements, or the like of constituents described in the embodiments may be appropriately changed according to the configurations, various conditions, or the like of apparatuses to which the invention is applied. Therefore, the sizes, materials, shapes, their relative arrangements, or the like of the constituents described in the embodiments do not intend to limit the scope of the invention to the following embodiments.

### First Embodiment

A liquid container according to the present invention can be applied to a printer, a copier, and a facsimile device as a liquid ejecting device. In this embodiment, the liquid container used for the liquid ejecting device of an inkjet recording method using ink as the liquid is explained as an example, but the liquid is not limited to ink in application of the present invention.

Hereinafter, a first embodiment of the present invention will be explained. FIG. **1** is a perspective view schematically illustrating a liquid ejecting device **100** in which a liquid container **1** according to the first embodiment is mounted. The liquid ejecting device **100** includes a recording head, an accommodating portion of a recording medium, a conveyance mechanism of the recording medium (these are not shown) and the like. In the liquid ejecting device **100**, the liquid container **1** as an ink pack accommodated in a cassette **2** is mounted. The cassette **2** is a case detachably attached to a main body of the liquid ejecting device **100**, and by attaching the cassette **2** to the liquid ejecting device **100** in a state in which the liquid container **1** is mounted, the liquid in the accommodating portion of the liquid container **1** can be supplied to the liquid ejecting device **100**.

In the following explanation and drawings, an X-direction, a Y-direction, and a Z-direction are defined as follows. The X-direction is a longitudinal direction of the liquid container **1**, in which the liquid container **1** and the cassette **2** are inserted into/removed from the liquid ejecting device **100**, when the liquid container **1** is attached to/detached from the liquid ejecting device **100**. The Y-direction is a width direction of the liquid container **1**, and the Z-direction is a thickness direction of the liquid container **1**. In this embodiment, the X-direction, the Y-direction, and the Z-direction are orthogonal to one another. In the following explanation, a state where the liquid container **1** is mounted in the liquid ejecting device **100** is called a mounted state. In the mounted state, the liquid container **1** is mounted in the liquid ejecting device **100** with the thickness direction in parallel to a vertical direction.

FIG. **1** is a perspective view of the liquid ejecting device **100** in which the liquid container **1** according to the first embodiment is mounted. The liquid ejecting device **100** includes a recording head, an accommodating portion of a recording medium, a conveyance mechanism of the recording medium (these are not shown) and the like, and the cassette **2** is provided capable of attachment/detachment. When the cassette **2** is inserted into and attached to the liquid ejecting device **100** in the X-direction in the state where the liquid container **1** is accommodated in the cassette **2**, the liquid container **1** is mounted in the liquid ejecting device **100**.

The liquid container **1** accommodates ink to be ejected from the recording head of the liquid ejecting device **100**. In this embodiment, four pieces of the liquid containers **1** accommodating ink in each of colors cyan (C), magenta (M), yellow (Y), and black (K) are accommodated in the cassettes **2**, respectively, and mounted in the liquid ejecting device **100**. The four cassettes **2** are attached to the liquid ejecting device **100** adjacently in the width direction (Y-direction) of the liquid container **1**. Sizes of the four liquid containers **1** of this embodiment are the same, but in application of the present invention, the liquid container **1** for the black ink, for example, may be made larger than the liquid containers **1** for the ink in the other colors. Moreover, the present invention can be also applied to a monochrome printer in which the liquid container for the ink only in one color is mounted or the like.

FIG. **2** is a perspective view of the liquid container **1** attached to the cassette **2**. The liquid container **1** of this embodiment is constituted by a bag member **10** accommodating ink inside and an adaptor **20** positioned with respect to the cassette **2**. The bag member **10** is fixed to the adaptor **20**, and the bag member **10** and the adaptor are provided adjacently in a longitudinal direction of the liquid container **1**. When the cassette **2** is attached to the liquid ejecting device **100** in a state where the adaptor **20** is positioned with respect to the cassette **2**, the liquid container **1** is connected to the liquid ejecting device **100**, and the liquid in the bag member **10** can be supplied to the liquid ejecting device **100**. That is, the adaptor **20** is a member for improving handling when the liquid container **1** is connected to the cassette **2** and the liquid ejecting device **100** and is a positioning member which is positioned with respect to the liquid ejecting device **100**.

The bag member **10** has a bag-shaped accommodating portion **10***a* having flexibility, a supply port **10***b* provided at an end portion in the longitudinal direction of the accommodating portion **10***a*, and a handle portion **10***c* for an operator to grasp. Ink in which a pigment as a sedimentation component is diffused in a solvent is accommodated in the accommodating portion **10***a* of this embodiment, but in application of the present invention, types of the liquids to be accommodated in the accommodating portion **10***a* are not limited to that. The accommodating portion **10***a* shown in FIG. **2** is in a state not accommodating the ink, and when the ink is filled, the accommodating portion **10***a* swells.

The ink accommodated in the accommodating portion **10***a* is supplied to the liquid ejecting device **100** through the supply port **10***b*. The supply port **10***b* extends in the longitudinal direction (X-direction) of the liquid container **1**, that is, an attachment/detachment direction of the liquid container **1** and the cassette **2** with respect to the liquid ejecting device **100**, and a supply direction of the ink from the bag member to the liquid ejecting device **100** is in parallel to the attachment/detachment direction.

In the bag member **10**, a first end portion **10***a***1** in which the supply port **10***b* of the accommodating portion **10***a* is provided in the longitudinal direction is connected to the adaptor **20**. A second end portion **10***a***2** on a side opposite to the first end portion is not fixed to the adaptor **20**. The handle portion **10***c* extends from the first end portion **10***a***1** side to the second end portion **10***a***2** side of the accommodating portion **10***a* by opposing the accommodating portion **10***a*. Since the operator can grip the handle portion **10***c* for the operation when mounting the liquid container **1** on the cassette **2** or removing it from the cassette **2**, workability is favorable.

Moreover, the handle portion **10***c* extends in a direction gradually separated from the accommodating portion **10***a* as it gets closer to the second end portion **10***a***2** from the first end portion **10***a***1** in the longitudinal direction of the liquid container **1**. That is, the handle portion **10***c* is disposed so that one end is fixed to the first end portion **10***a***1** of the accommodating portion **10***a*, and a clearance is given with respect to the accommodating portion **10***a*. By having such a constitution that the handle portion **10***c* has the clearance with respect to the accommodating portion **10***a* as above, a hand can be inserted into a space between the handle portion **10***c* and the accommodating portion **10***a*, and the handle portion **10***c* can be gripped easily.

The adaptor **20** has an electric connection portion (not shown) to be electrically connected to the liquid ejecting device **100**. When the liquid container **1** is attached together with the cassette **2** to the liquid ejecting device **100**, the adaptor **20** is electrically connected to the liquid ejecting device **100**, and the ink can be supplied from the bag member **10** to the liquid ejecting device **100**.

FIG. **3** is an exploded perspective view illustrating a constitution of the bag member **10**. The accommodating portion **10***a* of this embodiment is a pillow-type bag formed by placing an upper film **12** over a lower film **13**, each having a substantially rectangular shape, so that they are welded and bonded to each other on a welding portion **15** on opposing surfaces thereof. The welding portion **15** as a bonding region is outer peripheral portions of the upper film **12** and the lower film **13** excluding a portion constituting the handle portion **10***c*, and an accommodating region surrounded by the welding portion **15** forms an accommodating space for accommodating the ink. In FIG. **3**, in order to explicitly indicate the region of the welding portion **15**, the welding portion **15** is shown with hatching. The shape of the accommodating portion of the bag member may be either one of the pillow type and a gazette type with gore, and the shape thereof is not limited as long as the liquid such as ink can be accommodated and can be contained in the cassette **2**. Moreover, the bag member **10** of this embodiment is constituted by two pieces of sheet-state members, that is, the lower film **13** as a first sheet-shaped member and the upper film **12** as a second sheet-shaped member, but application of the present invention is not limited to that. For example, it may be such a constitution that one piece of the sheet-shaped member is folded, and the outer peripheral portion is bonded so that the accommodating space for the liquid is formed inside.

The lower film **13** as the first sheet-shaped member and the upper film **12** as the second sheet-shaped member constituting the bag member **10** only need to be formed of a material having flexibility and gas-barrier characteristics. The material used for each of the sheet-shaped members includes polyethylene terephthalate (PET), nylon, polyethylene, and the like, for example. Moreover, the upper film **12** and the lower film **13** may be a film with a laminated structure in which a plurality of films constituted by these materials are laminated. When the laminated structure is adopted, such a constitution is considered that an outer layer is formed of PET or nylon, which is excellent in impact resistance, and an inner layer is formed of polyethylene, which is excellent in ink resistance, for example. In addition, as a layer included in the laminated structure, a layer on which aluminum or the like is vapor-deposited may be included.

The bag member **10** has a spout portion **14** as a supply port member having the supply port **10***b* bonded between the two sheet-shaped members, that is, the upper film **12** and the lower film **13**. That is, the bag member **10** is manufactured such that a welding portion **15** is welded in a state where the spout portion **14** is sandwiched by the upper film **12** and the lower film **13**. At this time, the spout portion **14** is sandwiched by the upper film **12** and the lower film **13** so that the provided supply port **10***b* communicates between the accommodating space and an outside of the bag member **10**. Moreover, two spout holes **16** provided in the spout portion **14** and two film holes **17** provided in each of the upper film **12** and the lower film **13** are overlapped and penetrate in a thickness direction of the bag member **10**. The bag member **10** is fixed to the adaptor **20** by the spout hole **16** and the film hole **17** provided in the bag member **10**. A diameter of the film hole **17** only needs to be formed larger than the diameter of the spout hole **16** and only needs to be in a mode not preventing engagement between the bag member **10** and the adaptor **20**. Moreover, the spout portion **14** in this embodiment is formed of a plastic material, but the material of the spout portion **14** is not limited to that.

The ink supply into the accommodating portion **10***a* formed by the upper film **12** and the lower film **13** is performed after the upper film **12** and the lower film **13** are welded. The supply port **10***b* of the spout portion **14** has a valve provided so that the ink flows only in a direction of supply from the accommodating portion **10***a* to the liquid ejecting device **100**. Thus, when the ink is to be injected into the accommodating portion **10***a*, the ink is injected through an injection port (not shown) provided in the spout portion **14**, and the injection port is sealed after that.

From the first end portion **10***a***1** side of the lower film **13**, the handle portion **10***c* extends in the longitudinal direction of the bag member **10**, and the handle portion **10***c* is provided on the lower film **13** integrally. The lower film **13** is a flat sheet of film, and in a state not connected to the adaptor **20**, the handle portion **10***c* extends from the first end portion **10***a***1** of the accommodating portion **10***a* in a direction gradually separated from the accommodating portion **10***a*. The handle portion **10***c* has an opening having a rectangular shape opened in the thickness direction formed in the rectangular film, and the shape is easily gripped by the operator. The shape of the handle portion **10***c* is not limited to that and may be a shape with a rounded corner portion or a shape without a hole formed, for example. Moreover, the handle portion **10***c* in this embodiment is provided on the lower film **13**, but it may be so constituted that the handle portion **10***c* is provided on the upper film **12**.

FIG. **4**A is an exploded perspective view illustrating a constitution of the liquid container **1**, and FIG. **4**B is an exploded rear view illustrating the constitution of the liquid container **1** when viewed from a side opposite to the supply port side. The adaptor **20** as the positioning member is constituted by two members, that is, an upper adaptor **21** and a lower adaptor **22**.

In an adjacent direction of the bag member **10** and the adaptor **20** (longitudinal direction of the liquid container **1**), two depressed portions **21***a* recessed from the end portion on the bag member side to the adaptor side are provided on the upper adaptor **21**. The depressed portion **21***a* of this embodiment has a semicircular shape when viewed from the thickness direction and has a groove shape extending by penetrating in the thickness direction. Moreover, two depressed portions **22***a* are provided similarly on the lower adaptor **22**, too. When the upper adaptor **21** and the lower adaptor **22** are engaged, the depressed portions **21***a* and **22***a* are disposed by overlapping when viewed from the thickness direction, and the depressed portions **21***a* and **22***a* are used for positioning of the liquid container **1** with respect to the cassette **2**. Moreover, the upper adaptor **21** and the lower adaptor **22** of this embodiment are formed of a plastic material, but the material is not limited to that.

The bag member **10** is fixed to the adaptor **20** by being sandwiched by the upper adaptor **21** and the lower adaptor **22**. At this time, in a state where the lower film **13** is folded so that the handle portion **10***c* is opposed to a surface of the upper film **12** forming an outer surface of the accommodating portion **10***a*, the bag member **10** is sandwiched by the upper adaptor **21** and the lower adaptor **22**. In other words, the handle portion **10***c* is provided on the lower film **13** by being contiguous with the accommodating portion **10***a* through a folded portion **10***e*.

The bag member **10** is fixed to the adaptor **20** by engagement between an engagement hole **10***d* constituted by the spout hole **16** and the film hole **17** and a protruding portion **22***b* provided on the lower adaptor **22** and extending in the thickness direction. In a state where the handle portion **10***c* of the bag member **10** is folded, the protruding portion **22***b* is inserted through the spout hole **16** and the film hole **17**, and the bag member **10** is sandwiched by the upper adaptor **21** and the lower adaptor **22** so that the liquid container **1** is formed. Moreover, the upper adaptor **21** and the lower adaptor **22** are connected to each other by engagement between an engaging portion and an engaged portion, not shown. The engaging portion and the engaged portion only need to be a claw portion and a depressed portion in which the claw portion is fitted or the like, for example.

FIG. **5**A is a top view illustrating peripheral portions of the adaptor **20** and the handle portion **10***c* of the liquid container **1**. FIG. **5**B is an A-A sectional view of FIG. and illustrates a detailed constitution of the handle portion **10***c*. FIG. **5**C is a sectional view illustrating a detailed constitution of the adaptor **20** excluding the bag member **10** from the FIG. **5**B.

In the adaptor **20**, a depressed portion **20***a* constituted by the depressed portion **21***a* of the upper adaptor **21** and the depressed portion **22***a* of the lower adaptor **22** is provided. In the ink supply direction from the liquid container **1** to the liquid ejecting device **100** (longitudinal direction of the liquid container **1**), the depressed portion **20***a* is provided in the end portion on an upstream side of the adaptor **20** and has a groove shape extending in a direction perpendicular to the thickness direction.

Moreover, as shown in FIG. **5**C, in the adaptor **20**, a clearance portion **20***b* as a space in which the folded portion **10***e* of the bag member **10** is disposed is formed between the upper adaptor **21** and the lower adaptor **22**. As shown in FIG. **5**B, when the bag member **10** is engaged with the adaptor **20**, parts of the folded portion **10***e* and the handle portion **10***c* of the bag member **10** are positioned in the clearance portion **20***b*.

As a surface for forming the clearance portion **20***b*, the adaptor **20** has an upper surface **20***b***1** and a lower surface **20***b***2** as opposing surfaces opposing so as to sandwich the folded portion **10***e* and a region in the vicinity thereof in an opposing direction of the accommodating portion **10***a* and the handle portion **10***c*. The upper surface **20***b***1** is a surface in contact with the region in the vicinity of the folded portion **10***e* of the handle portion **10***c*, and the lower surface **20***b***2** is a surface in contact with the region in the vicinity of the folded portion **10***e* of the lower film **13**. In this embodiment, the lower surface **20***b***2** extends horizontally, and the upper surface **20***b***1** is inclined by an inclination angle θ with respect to the lower surface **20***b***2**. And an opposing gap between the upper surface **20***b***1** and the lower surface **20***b***2** becomes wider as it goes from the first end portion **10***a***1** to the second end portion **10***a***2**. The handle portion **10***c* is urged in a direction brought into contact with the upper surface **20***b***1** by a force (restoring force of the material constituting the bag member **10**) of the folded portion **10***e* of the bag member **10** to return to straight and thus, it is disposed so as to follow the upper surface **20***b***1**, and a folding-back angle of the handle portion **10***c* is regulated by the inclination angle θ. By constituting as above, when the bag member **10** is fixed to the adaptor **20**, such a posture can be maintained that the handle portion **10***c* extends so as to gradually separate from the accommodating portion **10***a* as it goes from the first end portion **10***a***1** side to the second end portion **10***a***2** side. Therefore, by disposing the folded portion **10***e* in the clearance portion **20***b*, the handle portion **10***c* is not crushed by the upper adaptor **21** and the lower adaptor **22** but the handle portion **10***c* is provided so as to have a gap with respect to the accommodating portion **10***a*. That is, since the handle portion **10***c* is disposed so as to slightly float in the thickness direction (Z-direction) with respect to the accommodating portion **10***a*, the operator can insert the hand between the handle portion **10***c* and the accommodating portion **10***a* and grip the handle portion **10***c* easily, which can realize the constitution with excellent handling performance.

As described above, the clearance portion **20***b* of the adaptor **20** is formed in conformance with the folded portion **10***e* of the bag member **10**. As shown in FIG. **5**C, when seen on a section orthogonal to a width direction (Y-direction) of the liquid container **1**, a sectional shape of the clearance portion **20***b* is substantially a trapezoid, and a width of the clearance portion **20***b* in the thickness direction of the bag member **10** becomes larger as it gets closer to an opening side (the bag member **10** side). That is, assuming that the width on the opening side of the clearance portion **20***b* is D1, and a width on a depth side farthest from the opening is D2, the sectional shape of the clearance portion **20***b* is preferably a trapezoid of D1>D2. In this embodiment, the sectional shape of the clearance portion **20***b* is a trapezoid by considering manufacture difficulty and functionality, but it may be a triangle of D2=0, and the surface on which the clearance portion **20***b* is formed does not necessarily have to be a plane.

Moreover, when the handle portion **10***c* is excessively long or is excessively separated from the accommodating portion **10***a*, there is a possibility that the handle portion **10***c* protrudes from the cassette **2**, interferes at attachment of the liquid ejecting device **100**, and causes nonconformity. Thus, assuming that a length of the handle portion **10***c* is L, a height from the lower surface **20***b***2** of the clearance portion **20***b* (insertion position of the bag member **10**) to an upper end of the upper adaptor **21** is H, and the inclination angle of the clearance portion is θ, θ is preferably smaller than $\sin^{-1}$ (H/L). By constituting them so as to satisfy $\theta<\sin^{-1}$ (H/L) as above, the end portion of the handle portion **10***c* does not protrude from the upper surface of the upper adaptor **21**, and the interference of the handle portion **10***c* when the liquid container **1** is mounted on the cassette **2** and is inserted into/removed from the liquid ejecting device **100** can be avoided.

FIG. **6** is a top view partially illustrating a state where the liquid container **1** is mounted on the cassette **2**. The liquid container **1** is mounted on the cassette **2** so that the handle portion **10***c* of the bag member **10** is directed to an upper side in the vertical direction and to the opening side of the cassette **2**. Moreover, the liquid container **1** is mounted on the cassette **2** so that a contact surface of the liquid container **1** with respect to the cassette **2** is perpendicular to the thickness direction of the liquid container **1**. Therefore, when the liquid container **1** is mounted on the cassette **2** or when the liquid container **1** is removed from the cassette **2**, the operator can perform setting on the cassette **2** while gripping the handle portion **10***c*, which is favorable in operability. In this constitution, the cassette **2** has two positioned portions **2***a* each having a columnar shape extending so as to protrude in the thickness direction (Z-direction) from a bottom surface, and the liquid container **1** is accommodated in a portion surrounded by an inner wall **2***b*. The positioning in the width direction (Y-direction) of the liquid container **1** with respect to the cassette **2** is performed by engagement between the depressed portion **20***a* of the liquid container **1** and the positioned portion **2***a* of the cassette **2**. Moreover, the positioning in the longitudinal direction (X-direction) of the liquid container **1** with respect to the cassette **2** is performed by sandwiching of the adaptor **20** between the positioned portion **2***a* and the inner wall **2***b* of the cassette **2**. In this embodiment, since the depressed portion **20***a* is formed having a semicircular shape on the end portion of the adaptor **20**, as compared with a case where the positioning portion is formed having a circular shape in the vicinity of a center portion, the adaptor can be constituted smaller, which results in space saving. The constitutions of the positioning portion and the positioned portion are not limited to that, and the sectional shapes of the positioning portion and the positioned portion do not have to be circular but may be polygonal, for example.

When the ink in the bag member **10** becomes empty, the cassette **2** is pulled out of the liquid ejecting device **100**, and the liquid container **1** is removed from the cassette **2**. In the removing work of the liquid container **1**, the operator can perform the work while gripping the handle portion **10***c* extending in the thickness direction, which is favorable in operability. As the new liquid container **1** to be attached to the cassette **2**, by disassembling the adaptor **20**, only the bag member **10** replaced with the new bag member **10** into which the ink was injected may be used or the new liquid container **1** which was replaced with the whole adaptor **20** may be used.

As described above, by means of the constitution of this embodiment, by forming the handle portion **10***c* on the bag member **10** so as to be integrated with the accommodating portion **10***a*, as compared with a case in which the handle portion is provided on the adaptor **20**, the liquid container **1** can be manufactured with a simple constitution. That is, since there is no need to manufacture the adaptor **20** of the liquid container **1** by combining a large number of molded products, a manufacturing cost can be reduced. In addition, since size reduction of the adaptor **20** of the liquid container **1** can be promoted, an increase in a size of the accommodating portion **10***a* can be expected and thus, improvement of accommodation efficiency of the ink can be promoted.

## Second Embodiment

Subsequently, a second embodiment according to the present invention will be explained. The second embodiment is different from the first embodiment in a point that the handle portion is constituted by an upper film **32** and a lower film **13**. In the following explanation, portions different from the first embodiment will be mainly explained, while the same or similar members as those in the first embodiment are given the same signs, and the explanation will be omitted.

FIG. **7** is an exploded perspective view of the liquid container **1** according to the second embodiment. The adaptor **20** provided in the liquid container **1** of this embodiment is constituted by the upper adaptor **21** and the lower adaptor **22** similarly to the first embodiment. A bag member **30** provided in the liquid container **1** is constituted by an upper film **32**, a lower film **13**, and the spout portion **14**, and a shape of the upper film **32** is different from that in the first embodiment.

The upper film **32** in this embodiment has the same shape as that of the lower film **13**, and a first handle forming portion and a second handle forming portion constituting the handle portion **10***c* are formed as the upper film **32** and the lower film **13**, respectively. That is, the handle portion of this embodiment is constituted by overlapping the two sheet-shaped members and bonding them to each other. By constituting as above, rigidity of the handle portion can be improved as compared with the first embodiment, and handling performance at attachment to/detachment from the cassette **2** can be improved. This embodiment is particularly effective when the rigidity of the handle portion is to be improved in such a case that a material used for the sheet-shaped member is excessively soft, and a gap cannot be provided between the handle portion and the accommodating portion, even if the handle portion is formed of one sheet of film, for example. Moreover, according to this embodiment, since the upper film **32** and the lower film **13** have the same shape, the number of components to be manufactured can be reduced, and ease of manufacture increases.

## Third Embodiment

Subsequently, a third embodiment according to the present invention will be explained. The third embodiment is different from the first embodiment in a point that it is so constituted that a lower film **43** is folded so that the handle portion is doubled. In the following explanation, portions different from those in the first embodiment will be mainly explained, while the same or similar members as those in the first embodiment are given the same signs, and the explanation will be omitted.

FIG. **8** is a perspective view of the lower film **43** according to the third embodiment. A portion constituting the handle portion of the lower film **43** of this embodiment includes a base portion **43***a*, a folded portion **43***b* protruding in a width direction from both ends in the width direction with respect to the base portion **43***a* and a folding line **43***c*, which is a boundary between the base portion **43***a* and the folded portion **43***b* and extends in a longitudinal direction. In this embodiment, for the purpose of improvement of rigidity of the handle portion **10***c*, the folded portion **43***b* is folded with respect to the base portion **43***a* with the folding line **43***c* as a base line, and the base portion **43***a* and the folded portion **43***b* are bonded. In this embodiment, the folded and overlapped base portion **43***a* and the folded portion **43***b* become a folded-back portion which is folded back to the accommodating portion **10***a*, and the folded-back portion constitutes the handle portion **10***c*. By constituting as above, the rigidity of the handle portion can be improved similarly to the second embodiment, as compared with the first embodiment. Here, the constitution of the lower film **43** was explained, but the upper film, instead of the lower film **43**, can also have the constitution similar to the lower film **43**.

Application of the present invention is not limited to the embodiments described above, but various changes can be made. For example, it may be so constituted that the handle portion has threefold or fourfold sheet-shaped members by combining the second embodiment and the third embodiment. Moreover, in the above embodiment, the bag member was manufactured by using a film in which the handle portion and the accommodating portion are integral, but simply if accommodation efficiency of the liquid is to be improved, the handle portion may be bonded to the accommodating portion by means of welding or the like. That is, by providing the handle portion so as to oppose the surface on a side opposite to the contact surface with respect to the cassette of the accommodating portion, the constitution can have excellent handling performance when the liquid container is to be removed from the cassette, while the accommodation efficiency of the liquid is improved, as compared with the case where the handle portion is provided on the adaptor.

Moreover, another variation to which the present invention is applied will be explained by referring to FIGS. **9**A and **9**B. FIGS. **9**A and **9**B are perspective views illustrating a liquid container according to the variation. As shown in FIG. **9**A, it may be so constituted as another variation that a handle portion **10***g* is provided on the end portion on a side opposite to the supply port **10***b* in the longitudinal direction of the bag member **10**, for example. In the case of the constitution as above, the handle portion **10***g* is provided with a gap with respect to the accommodating portion **10***a* by giving a strong bending tendency or by providing a bonding member so that the folded portion does not return to a straight posture.

Alternatively, as shown in FIG. **9**B, it may be so constituted that a handle portion **10***h* is provided across the width direction in the vicinity of the center in the longitudinal direction of the accommodating portion **10***a*. Such a constitution can be formed by forming a portion to become the handle portion **10***h* on either one of the upper film **12** and the lower film **13** in advance and by folding it so that a distal end is bonded to the accommodating portion. As described above, various changes can be made within a range not losing the identity with the invention embodied in the above embodiments, and in any case, a mass of the connecting member such as the adaptor is minimized, and the accommodation efficiency of the liquid can be improved.

According to the present invention, the liquid container with high liquid accommodation efficiency can be provided with a simple constitution.

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of Japanese Patent Application No. 2022-088866, filed on May 31, 2022, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

**1**. A liquid container being able to be detachably attached to a case which is detachably attached to a liquid ejecting device and supplying a liquid to the liquid ejecting device, the liquid container comprising:

a bag-shaped liquid accommodating portion for containing a liquid, the liquid accommodating portion being configured by overlapping two sheet materials and sealing a peripheral edge of the two sheet materials, the two sheet materials including a first sheet material and a second sheet material;

a handle portion adjacent to the liquid accommodating portion and including an opening formed by cutting out a part of a region not containing liquid of the second sheet material;

a supply portion that is provided at one end of the liquid accommodating portion and that is provided so as to be capable of leading liquid from inside of the liquid accommodating portion to outside of the liquid accommodating portion, the supply portion being (1) sandwiched between the first sheet material and a portion of the second sheet material that forms one side edge of the opening, and (2) sealed between the two sheet materials; and

an adapter that sandwiches and fixes the supply portion, the liquid accommodating portion, and a part of the handle portion in a state where the handle portion is folded back toward the liquid accommodating portion at a portion where the supply portion is disposed, and that positions the liquid accommodating portion and the supply portion with respect to the case,

wherein the liquid accommodating portion includes an engaging portion engaged with the adapter.

**2**. The liquid container according to claim **1**, wherein the handle portion extends so as to be gradually separated from the liquid accommodating portion as the handle portion extends from the one end of the liquid accommodating portion portion to toward another end of the liquid accommodating portion.

**3**. The liquid container according to claim **2**, wherein the adapter includes two members facing each other so as to sandwich the supply portion, the liquid accommodating portion, and a part of the handle portion, and includes an inclined surface that defines a folding-back angle of the handle portion.

**4**. The liquid container according to claim **1**, wherein the adapter includes an engaging portion that engages with an engaged portion of the case.

**5**. A liquid container being able to be detachably attached to a case which is detachably attached to a liquid ejecting device and supplying a liquid to the liquid ejecting device, the liquid container comprising:

a bag-shaped liquid accommodating portion for containing a liquid, the liquid accommodating portion being configured by folding a sheet material and sealing a peripheral edge of the sheet material;

a handle portion adjacent to the liquid accommodating portion and including an opening formed by cutting out a part of a region not containing liquid of the sheet material;

a supply portion that is provided at one end of the liquid accommodating portion and that is provided so as to be capable of leading liquid from inside of the liquid accommodating portion to outside of the liquid accommodating portion, the supply portion being sandwiched and sealed between (1) a first portion of the sheet material and (2) a second portion of the sheet material that forms one side edge of the opening; and

an adapter that sandwiches and fixes the supply portion, the liquid accommodating portion, and a part of the handle portion in a state where the handle portion is folded back toward the liquid accommodating portion at a portion where the supply portion is disposed, and that positions the liquid accommodating portion and the supply portion with respect to the case,

wherein the liquid accommodating portion includes an engaging portion engaged with the adapter.

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